Answer key: Lesson 4, The 100 Square Explorer

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Colour 10, 20, 30 and so on to 100 as you count in 10s. Fill in the blanks.

Answer: They end in 0. They make a line going down.

The numbers are 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. Every one ends in 0. Each is 10 more than the one before, which is one row down, so they make a straight line down the last column.

Watch for: Colouring every number in the last row (91 to 100).; Saying the line goes across: the colours are one under the other, so it goes down.

S2. Count in 5s. Colour 5, 10, 15 and keep going to 100. Fill in the blanks.

Answer: They end in 5 or 0. They make 2 lines going down.

The numbers are 5, 10, 15, 20, and on to 100. They end in 5 or 0. The 5s sit in the fifth column and the 10s in the last column, so there are two lines going down.

Watch for: Skipping a number, such as 35 or 45.; Saying 1 line because the tens are in a line too (there are 2 lines: the 5s and the 0s).

S3. What are the next two numbers in each of these patterns?

Answer: 10, 12 | 30, 35 | 60, 70 | 41, 51

(a) Counting in 2s: 8, then 10, 12. (b) Counting in 5s: 25, then 30, 35. (c) Counting in 10s: 50, then 60, 70. (d) Each step is 10 more, one row down: 31, then 41, 51.

Watch for: (d) writing 32 and 33: the jump is 10, not 1.; Changing the jump half way along.

S4. Count in 2s. Colour 2, 4, 6 and so on to 30. Look at the last digits.

Answer: 2, 4, 6, 8, 0 (any order)

The coloured numbers are 2, 4, 6, 8, 10, 12 and on to 30. Their last digits are 2, 4, 6, 8, 0, and then the same five again. The colours make stripes.

Watch for: Colouring 1, 3, 5 as well.; Writing 5 or 1 as a last digit.

S5. Fill in the missing number in each empty square.

Answer: a 34, b 43, c 55

(a) 34 is between 33 and 35. (b) 43 is under 33, which is 10 more than 33, and above 53. (c) 55 is under 45 (10 more) and between 54 and 56.

Watch for: Writing 44 for b: 44 is next to 43, not under 33.; Adding 1 when moving down instead of 10.

S6. Siti counts in 5s: 5, 10, 15, 20, 26, 30, 35. One number is wrong. Circle it.

Answer: 26 is wrong. She should have said 25. Counting in 5s ends in 0 or 5.

Counting in fives, 20 is followed by 25, then 30. 26 ends in 6, but numbers counted in fives end in 5 or 0. So Siti should have said 25.

Watch for: Circling 30 or 35.; Writing 25 but giving no reason (a clear sentence about the last digit is enough).

Think harder (page 3)

T1. Put a counter on 27 on your 100 square. Move the counter down 3 rows. Which number does it land on? Write how you know on the lines.

Answer: 57 (each row down is 10 more, so 3 rows is 30 more)

Each row down adds 10, so 3 rows add 30. 27 + 30 = 57. Check on the square: 37, 47, 57.

Watch for: Writing 30 or 33 (adding 3 instead of 3 tens).; Counting 3 squares along the row.

T2. A counter is on 14. It moves down 2 rows. Then it moves right 3 squares. Which number does it land on? Write each step on the lines.

Answer: 37 (14, 24, 34, then 35, 36, 37)

Down 2 rows is 20 more: 14, 24, 34. Right 3 squares is 3 more: 35, 36, 37. It stops on 37.

Watch for: Adding 2 and 3 to 14 (19).; Going down by 2 squares instead of 2 rows (16).

T3. Look at this pattern: 6, 7, 9, 12, 16, ___, ___. First write the gaps between the numbers. Then write the next two numbers.

Answer: The gaps are 1, 2, 3, 4. The next numbers are 21 and 27.

The gaps are 1, 2, 3, 4: each gap is 1 bigger. The next gap is 5, so 16 + 5 = 21. The gap after is 6, so 21 + 6 = 27.

Watch for: Adding 4 each time (20, 24).; Finding the gaps but forgetting to use them to go on.

T4. Mei counts in 2s up to 30: 2, 4, 6, 8 and so on. Ali counts in 5s up to 30: 5, 10, 15, 20 and so on. Write the numbers they both say.

Answer: 10, 20 and 30

Mei says 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30. Ali says 5, 10, 15, 20, 25, 30. Both say 10, 20 and 30.

Watch for: Writing only 10.; Including 5, 15 or 25: Mei does not say numbers ending in 5.

Go further (page 4)

G1. Here is a number chain: 2, 6, _ _ _ _ _ _ _, 14, 22, _ _ _ _ _ _ _. After 2 and 6, each number is the last two numbers added together.

Answer: 8 and 36 (2 + 6 = 8, and 14 + 22 = 36)

2 + 6 = 8, so the first blank is 8. Check: 6 + 8 = 14 and 8 + 14 = 22, which are in the chain. The last blank is 14 + 22 = 36.

Watch for: Adding 6 each time (8, 14, 20).; Filling the first blank but not the last.

G2. Two patterns are mixed in one list: 5, 50, 10, 45, 15, 40, _ _ _ _ _ _ _, _ _ _ _ _ _ _. Write the next two numbers. Then write the 10th number in the list.

Answer: The next two numbers are 20 and 35. The 10th number is 30.

Numbers 1, 3, 5 go up by 5: 5, 10, 15, then 20. Numbers 2, 4, 6 go down by 5: 50, 45, 40, then 35. The list goes on 20, 35, 25, 30. The 10th number is 30.

Watch for: Treating the list as one pattern (adding or taking away the same amount).; Getting the 8th number right and the 10th wrong.

G3. Look at this block of 4 squares.

Answer: Both add up to 79. It always works: 1 and 10 are added in both pairs.

If the top left number is a, the others are a + 1, a + 10 and a + 11. One slant is a + (a + 11) and the other is (a + 1) + (a + 10). Both are the same: 2 a and 11. For the first block both are 79.

Watch for: Only trying one block and saying it always works.; Adding the numbers across the rows instead of along the slants.

Investigation (page 5): Counting in 3s

Counting in 3s makes slanting lines going down and to the left, not a straight column, because a row has 10 squares and 10 is not said when we count in 3s. The picture repeats every 3 rows: 33, 36 and 39 sit under 3, 6 and 9.

Look for: A child who colours carefully, says what shape they see and gives a reason that uses rows of ten. Any clear noticing counts (zig-zag, slanting, it moves left each row). Colouring to 60 shows the repeat. Do not name the word multiple.

Exit ticket (page 5)

E1. Count in 5s. What are the next two numbers?

Answer: 50, 55

Counting in fives: 45, then 50, then 55.

Watch for: Writing 46 and 47.; Counting in tens (55 and 65).

E2. A counter is on 26. First it moves down 1 square. Then it moves right 1 square.

Answer: 36, then 37

Down one row is 10 more: 36. Right one square is 1 more: 37.

Watch for: Writing 27 for down.; Writing 46 (adding 20).

E3. Mei counts in 2s: 2, 4, 6, 8 and so on. Will she say 25?

Answer: No. 25 ends in 5. Counting in 2s ends in 0, 2, 4, 6 or 8.

Counting in twos, the last digit is always 0, 2, 4, 6 or 8. 25 ends in 5, so Mei does not say 25.

Watch for: Saying right because 25 is next to 24 or 26.; Not giving a reason.

Sneak-peek puzzles (not in the workbook)

1. 25 and 30
2. 46. Down one row is 10 more, because a row has ten squares.